

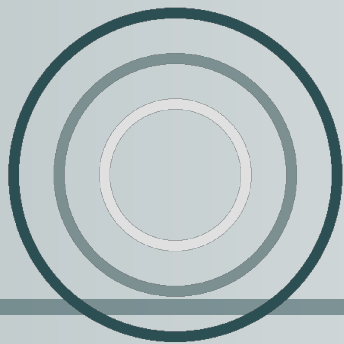
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Curso Maker

Módulo 2



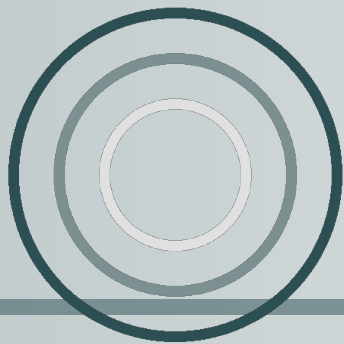
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PREFACE

This short book collects the definitions and first theorems of graph colouring, aimed at a reader who has seen basic graph theory but not yet chromatic numbers.

– Poggers Maker
2026



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PART I

FOUNDATIONS

The first part develops the basic vocabulary of graph colouring and proves the elementary upper bounds every later result builds on.



PROPER COLOURINGS

Part I

Sec 1.1 Definitions

Definition 1.1.1 (Proper colouring). A proper k -colouring of a graph $G = (V, E)$ is a function $c: V \rightarrow \{1, \dots, k\}$ such that $c(u) \neq c(v)$ whenever $uv \in E$.

Definition 1.1.2 (Chromatic number). The chromatic number $\chi(G)$ is the least k for which G admits a proper k -colouring.

Theorem 1.1.1 (Greedy bound). For any graph G with maximum degree Δ , $\chi(G) \leq \Delta + 1$.

Proof. Order the vertices arbitrarily and colour them in that order, at each step assigning the least colour not used by an already-coloured neighbour. Each vertex has at most Δ neighbours, so at most Δ colours are forbidden, leaving one of the first $\Delta + 1$ colours free. ■

Example 1.1 (An odd cycle). The cycle C_5 has $\Delta = 2$, so Theorem 1.1.1 gives $\chi(C_5) \leq 3$. In fact $\chi(C_5) = 3$: two colours are never enough for a cycle of odd length, since the colours would have to alternate and cannot close up consistently.

Lemma 1.1.1 (Bipartite graphs). A graph G is bipartite if and only if $\chi(G) \leq 2$.

Remark. Lemma 1.1.1 makes 2-colourability and bipartiteness the same question, and both can be checked in linear time by breadth-first search.

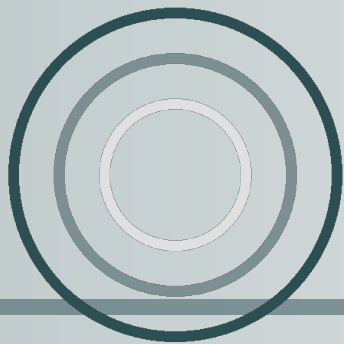


THE FOUR COLOUR THEOREM

Part I

Proposition 2.0.1 (Planar upper bound). *Every planar graph satisfies $\chi(G) \leq 4$.*

Proposition 2.0.1 is the Four Colour Theorem, first proved by computer-assisted case analysis by Appel and Haken (1977), and later given a more tractable proof by Robertson et al. (1997).



BIBLIOGRAPHY

Kenneth Appel and Wolfgang Haken. Every planar map is four colorable. *Illinois Journal of Mathematics*, 21(3):429–490, 1977.

Neil Robertson, Daniel Sanders, Paul Seymour, and Robin Thomas. The four-colour theorem. *Journal of Combinatorial Theory, Series B*, 70(1):2–44, 1997.